

From $\mathbb{Z}$ to the Multiverse: The View from $p = 7$

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Abstract

The set of integers $\mathbb{Z}$ is the unique infinite cyclic group up to isomorphism. The Bost–Connes (BC) algebra is the quantum statistical mechanical incarnation of $\mathbb{Z}$, whose partition function is precisely the Riemann ζ function. The Euler product factorization of the ζ function decomposes the BC algebra naturally into independent channels indexed by primes—each prime p corresponds to a complete physical channel with its own spacetime dimension, gauge group, and fermionic structure. These channels are not postulated; they are intrinsic structures within the BC algebra labeled by primes. We prove that $p = 7$ is the unique channel simultaneously possessing four-dimensional spacetime and $SU(2) \times SU(3)$ gauge group—the observed physics corresponds to the $p = 7$ channel.

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Claim labels. Throughout this paper, statements are marked with confidence labels: **[A]** rigorous mathematical theorem or algebraic derivation; **[A*]** numerically well-verified but with certain bridges yet to be made fully rigorous; **[B]** constructive physical assumption or model-dependent inference; **[C]** physical interpretation or speculative conjecture.

1 From $\mathbb{Z}$ to BC

$\mathbb{Z}$ (the additive group of integers) is the unique infinite cyclic group up to isomorphism. It has a single generator and a single operation (addition), with no internal structure, no parameters, and no choices.

Multiplication is not an independent operation—it is the endomorphism algebra of addition: $\text{End}(\mathbb{Z}, +) = \mathbb{Z}$, where each endomorphism $\varphi(n) = mn$ is precisely “multiplication by m ”. The distributive law $a(b + c) = ab + ac$ is not an extra axiom but an automatic consequence of the action of endomorphisms on the group. The fundamental theorem of arithmetic (every positive integer factors uniquely into primes) follows immediately.

Each step in the following chain is determined by known mathematical theorems:

1. **$\mathbb{Z}$ is unique:** there is only one infinite cyclic group up to isomorphism (classification theorem for cyclic groups).
2. **$\text{End}(\mathbb{Z}, +) = \mathbb{Z}$:** every additive endomorphism φ satisfies $\varphi(n) = \varphi(1) \cdot n$, completely determined by $\varphi(1) \in \mathbb{Z}$. Multiplication is the endomorphism of addition—not an independent operation, but the endomorphism ring of the additive structure.
3. **$\mathbb{N}^*$ is unique:** injective endomorphisms = multiplication by $m \geq 1$. The unique subsemigroup of injective elements in $\text{End}(\mathbb{Z})$.
4. **$\text{Aff}(\mathbb{Z}) = \mathbb{Z} \rtimes \mathbb{N}^*$ is unique:** $\mathbb{N}^*$ acts on $\mathbb{Z}$ by multiplication; the semidirect product is uniquely determined by the distributive law. It encodes addition (translations $T_a : n \mapsto n + a$) and multiplication (scalings $\mu_m : n \mapsto mn$) together with their intertwining relation $\mu_m \circ T_a = T_{ma} \circ \mu_m$.
5. **$C^*(\mathbb{Z} \rtimes \mathbb{N}^*)$ is unique:** the BC algebra is uniquely determined by the construction of Bost–Connes [6]—given the affine group $\mathbb{Z} \rtimes \mathbb{N}^*$ and its crossed relations, the C^* -algebra is unique.
6. **$H = \log n$ is unique:** the unique function satisfying complete additivity $H(mn) = H(m) + H(n)$ (energy is additive over independent systems), non-negativity $H \geq 0$ (energy is non-negative), and slow variation $H(n + 1) - H(n) \rightarrow 0$ (energy differences between adjacent states vanish) is $c \cdot \log n$ (Erdős–Kátaï theorem [25]; independent proof: Ostrowski’s theorem excludes p -adic valuations, leaving only the Archimedean one [26]).
7. **$Z(\beta) = \zeta(\beta)$ is unique:** given $H = \log n$, the partition function of the KMS_β state $Z(\beta) = \sum n^{-\beta} = \zeta(\beta)$ is uniquely determined by the Bost–Connes classification theorem [6].

Every step in this chain is determined by known theorems. Consequently, the structure of $\mathbb{Z}$ uniquely yields the BC system—including the Hilbert space $\ell^2(\mathbb{N}^+)$, the Hamiltonian $H = \log n$, the partition function $\zeta(\beta)$, the KMS phase transition at $\beta = 1$, and the Galois symmetry $\hat{\mathbb{Z}}^*$.

The subsequent sections take this BC system as starting point and explore the physical channel associated to each prime p .

Definition 1.1 ($\mathbb{Z}$ as the initial ring $[\mathbf{A}]$). *$\mathbb{Z}$ is the initial object in the category of commutative rings: for every commutative ring R there exists a unique ring homomorphism $\mathbb{Z} \rightarrow R$.*

1.1 Observation as Counting

Definition 1.2 (Arithmetic definition of observation $[\mathbf{A}]$). *In the BC system, **observation** is defined as the selection of a KMS_β state φ_β ($\beta > 0$). The three algebraic counterparts of observation are:*

- **Distinguishability:** for the extremal KMS_β states with $\beta > 1$, $\varphi_\beta(\mu_n \mu_n^*) = n^{-\beta} / \zeta(\beta)$. Since $n^{-\beta}$ is strictly decreasing in n for $\beta > 0$, one has $\varphi_\beta(\mu_n \mu_n^*) \neq \varphi_\beta(\mu_m \mu_m^*)$ whenever $n \neq m$;
- **Accumulability:** φ_β is a positive linear functional on $\mathcal{A}_{\text{BC}}$, supporting finite superpositions and limit operations;
- **Irreversibility:** μ_n is an isometry but not a unitary ($\mu_n^* \mu_n = 1 \neq \mu_n \mu_n^*$); multiplication by n is irreversible.

These properties follow directly from the definition of the BC algebra (Definition 2.1), without additional assumptions.

In this sense, $\mathbb{Z}$ is not merely a collection of numbers but the algebraic encoding of “the act of counting itself.”

2 The BC Algebra—Quantum Statistical Mechanics of $\mathbb{Z}$

2.1 The Bost–Connes System

Bost and Connes [6] constructed the following system. A complete treatment is given in [7, 11].

Definition 2.1 (BC algebra $[\mathbf{A}]$). *The **Bost–Connes C^* -algebra** is defined as*

$$\mathcal{A}_{\text{BC}} := C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^*,$$

where $C^*(\mathbb{Q}/\mathbb{Z})$ is the group C^* -algebra of the discrete group $\mathbb{Q}/\mathbb{Z}$, and $\mathbb{N}^*$ acts on it by multiplication. $\mathcal{A}_{\text{BC}}$ is generated by the following generators and relations:

- $e(r)$, $r \in \mathbb{Q}/\mathbb{Z}$: unitary elements satisfying $e(r)^* = e(-r)$, $e(r)e(s) = e(r+s)$;
- μ_n , $n \in \mathbb{N}^*$: isometries satisfying $\mu_n^* \mu_n = 1$, $\mu_n \mu_n^* = \frac{1}{n} \sum_{r: nr=0} e(r)$;
- Crossed relations: $\mu_n e(r) = e(nr) \mu_n$, $\mu_n^* e(r) = \frac{1}{n} \sum_{ns=r} e(s) \mu_n^*$.

Remark 2.2 (Semigroup crossed product structure). *Note that $\mathbb{N}^*$ is a multiplicative semigroup, not a group (elements have no multiplicative inverse, i.e., one cannot “divide by n ”). Strictly speaking, $\mathcal{A}_{\text{BC}}$ is a **semigroup crossed product** (of Hecke type), in which $\mathbb{N}^*$ generates the algebra through the irreversible action of the isometries μ_n . This differs from a group crossed product: μ_n is an isometry but not a unitary ($\mu_n^* \mu_n = 1 \neq \mu_n \mu_n^*$), reflecting the irreversibility of multiplication by n . For the general theory of semigroup crossed products arising in number theory, see Laca–Raeburn [18].*

Definition 2.3 (Hamiltonian). *The Hamiltonian of the BC system is defined on the natural basis $|n\rangle$ ($n \in \mathbb{N}^*$) by*

$$H|n\rangle = (\log n)|n\rangle.$$

The partition function is $Z(\beta) = \sum_{n=1}^{\infty} n^{-\beta} = \zeta(\beta)$, convergent for $\beta > 1$.

2.2 KMS States and Phase Transition

Definition 2.4 (KMS state [A]). *At inverse temperature β , a KMS_β state of $(\mathcal{A}_{\text{BC}}, \sigma_t)$ is a continuous linear functional φ such that for every $a, b \in \mathcal{A}_{\text{BC}}$ there exists a function $F_{a,b}$, bounded and continuous on the strip $0 < \Im(z) < \beta$ and holomorphic in its interior, satisfying*

$$F_{a,b}(t) = \varphi(a\sigma_t(b)), \quad F_{a,b}(t + i\beta) = \varphi(\sigma_t(b)a).$$

The modular flow σ_t is defined by $\sigma_t(\mu_n) = n^{it}\mu_n$, $\sigma_t(e(r)) = e(r)$.

Theorem 2.5 (BC phase transition [A], [6]). *KMS_β states are structured by inverse temperature β as follows:*

- (i) $\beta \leq 1$: *there exists a unique KMS_β state;*
- (ii) $\beta > 1$: *the set of extremal KMS_β states is homeomorphic to the coset space $\widehat{\mathbb{Z}}^* = \varprojlim (\mathbb{Z}/n\mathbb{Z})^*$, which has continuum cardinality;*
- (iii) $\beta \rightarrow \infty$ (ground state): *extremal KMS states are parametrized by orbits of $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$, in bijection with elements of $\widehat{\mathbb{Z}}^*$.*

A phase transition occurs at $\beta = 1$: the number of extremal states jumps from 1 to $|\widehat{\mathbb{Z}}^|$ (profinite cardinality).*

2.3 Physical Interpretation of the β -Axis

Definition 2.6 (β -axis [C]). $\beta \in (0, \infty)$ is the **inverse observation depth** (physically, the inverse temperature). By Theorem 2.5, the structure of KMS_β states is as follows:

- $\beta \leq 1$: *unique KMS_β state (quantum disordered phase);*
- $\beta = 1$: *phase transition point, the number of extremal states jumps from 1 to $|\widehat{\mathbb{Z}}^*|$ (critical point of spontaneous symmetry breaking);*
- $\beta > 1$: *multiple KMS_β states coexist, extremal states parametrized by $\widehat{\mathbb{Z}}^*$ (spontaneous symmetry breaking);*
- $\beta \rightarrow \infty$ (ground state): *extremal KMS states are parametrized by orbits of $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$; $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \cong \widehat{\mathbb{Z}}^*$ emerges in this manner.*

Physical interpretation [C]: $\beta = 0$ corresponds to the Haar measure (complete superposition, no observation); $\beta = 1$ is the phase transition, physically corresponding to the critical point of “emergence of observation”; for $\beta > 1$, “a channel has been selected.”

Remark 2.7 (Self-referential loop [C]). *The following closed loop provides physical intuition for the BC system:*

$$\text{Observation} \rightarrow \mathbb{Z} \rightarrow \mathcal{A}_{\text{BC}} \rightarrow \text{KMS} \xrightarrow{\beta=1} \text{Born rule} \rightarrow \text{Observation}.$$

Specifically:

- (i) *Observation produces counting; the algebraic model of counting is $\mathbb{Z}$;*

- (ii) The quantum statistical mechanization of $\mathbb{Z}$ yields $\mathcal{A}_{\text{BC}}$;
- (iii) The KMS state at $\beta = 1$ yields a probability measure (a candidate for the Born rule);
- (iv) The Born rule assigns definite outputs to observations, closing the loop.

Remark: the Born rule $P = |c_n|^2$ is uniquely determined by the Cauchy functional equation and non-negative pinching $[A]$ (see [1] for details). This chain is a physical interpretation, not a mathematical theorem.

3 Prime Channels—Each p Determines Complete Physics

3.1 Primes as Euler Factors

Theorem 3.1 (BC decomposition by primes $[A]$). *The partition function of the BC algebra decomposes as an Euler product:*

$$\zeta(\beta) = \prod_p \frac{1}{1 - p^{-\beta}}.$$

*Each prime p corresponds to an **Euler factor** (irreducible local factor) $\frac{1}{1-p^{-\beta}}$. The primes are the Euler factors of the partition function of the BC algebra.*

Remark 3.2 (Note on localization). *In the literature, the Euler factor at p is often associated to a local C^* -algebra. However, “the localization of $\mathcal{A}_{\text{BC}}$ at p , denoted $\mathcal{A}_{\text{BC},p}$,” is not a standard mathematical property of the BC algebra but a physical interpretation of this paper $[B]$ — $\mathcal{A}_{\text{BC}}$ as a C^* -algebra does not admit a direct localization; the above language is heuristic and should not be treated as a rigorous algebraic construction. The connection between local Euler factors and motives is studied in depth by Deninger [13]; for algebraic curves over finite fields, a similar Euler product decomposition of the Weil ζ function appears in [15].*

3.2 Spacetime Dimension: The Hermitian Matrix Argument

Theorem 3.3 (Minkowski structure of $\text{Herm}_2(\mathbb{C})$ $[A]$). *The space $\text{Herm}_2(\mathbb{C})$ of 2×2 Hermitian matrices has:*

- (i) real dimension 4;
- (ii) the determinant $\det : \text{Herm}_2(\mathbb{C}) \rightarrow \mathbb{R}$ is a non-degenerate quadratic form of signature $(1, 3)$:

$$\det \begin{pmatrix} t + z & x - iy \\ x + iy & t - z \end{pmatrix} = t^2 - x^2 - y^2 - z^2.$$

Therefore $\text{Herm}_2(\mathbb{C}) \cong \mathbb{R}^{1,3}$ (Minkowski space) as real quadratic spaces. Note: this step uses the continuous structure $\mathbb{R}^{1,3}$; in the discrete framework, “4-dimensional” follows directly from $|\text{Herm}_2| = 4$ (the number of free real parameters), without introducing Minkowski space.

Proof. The full derivation is given in [4] (the spacetime paper, Theorem 2.3). The core argument: $\text{Herm}_2(\mathbb{C})$ is parametrized by the Pauli matrices; the number of free real parameters equals 4; the determinant quadratic form has signature $(1, 3)$. In the discrete framework, “4-dimensional” follows directly from the parameter count, without introducing the continuous Minkowski space. $\square$

Remark 3.4 (The case of general a). *For general a , $\text{Herm}_a(\mathbb{C})$ has real dimension a^2 , but $\det$ is a polynomial of degree a , not a quadratic form. The case $a = 2$ is the unique instance where $\det$ is exactly a quadratic form, which is the mathematical reason why the Minkowski structure (Lorentz quadratic form) appears at $a = 2$ and nowhere else.*

Corollary 3.5 (Physical identification of four-dimensional spacetime [B]). *The CRT decomposition $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ for $p = 7$ yields the minimal factor $a = 2$. By Theorem 3.3, $\text{Herm}_2(\mathbb{C})$ is a four-dimensional real space whose determinant yields the Minkowski metric $(1, 3)$ [A]. The identification of $\text{Herm}_2(\mathbb{C})$ with spacetime—that is, identifying “the space of self-adjoint 2×2 matrices for the gauge group $\text{SU}(2)$ ” with the coordinate space of spacetime—is a **constructive assumption** [B], not a mathematical derivation.*

3.3 CRT Decomposition of the Gauge Group

Theorem 3.6 (CRT decomposition [A]). *Let $p > 2$ be a prime and $\varphi(p) = p - 1$. Let $p - 1 = \prod_i q_i^{a_i}$ be the prime factorization; then the Chinese Remainder Theorem yields a group isomorphism*

$$(\mathbb{Z}/p\mathbb{Z})^* \cong \bigoplus_i \mathbb{Z}/q_i^{a_i}\mathbb{Z}.$$

Theorem 3.7 (Factor-preserving unitary group [A]). *Let $V = \mathbb{C}^a \otimes \mathbb{C}^b$ ($a, b \geq 2$, $a \neq b$). Define the **factor-preserving unitary transformations** to be those $U \in U(ab)$ such that for all $A \in M_a(\mathbb{C})$ there exists $A' \in M_a(\mathbb{C})$ with $U(A \otimes I_b)U^* = A' \otimes I_b$, and for all $B \in M_b(\mathbb{C})$ there exists $B' \in M_b(\mathbb{C})$ with $U(I_a \otimes B)U^* = I_a \otimes B'$. Then the factor-preserving unitary group is isomorphic to*

$$G_{a,b} \cong (U(1) \times \text{SU}(a) \times \text{SU}(b)) / \mathbb{Z}_a \times \mathbb{Z}_b,$$

where $\mathbb{Z}_a \times \mathbb{Z}_b$ embeds via $(\omega, \omega') \mapsto ((\omega\omega')^{-1}, \omega I_a, \omega' I_b)$.

Proof. The complete five-step proof (Skolem–Noether $\rightarrow$ double commutant $\rightarrow$ quotient group $\rightarrow$ kernel computation) is given in [2] (the four-forces unification paper, Theorem 3.7). $\square$

Corollary 3.8 (Gauge group at $p = 7$ [A]). *For $p = 7$: $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ (Theorem 3.6). The CRT decomposition of the group algebra yields a **natural** tensor product structure:*

$$\mathbb{C}[(\mathbb{Z}/7\mathbb{Z})^*] \cong \mathbb{C}[\mathbb{Z}/2\mathbb{Z}] \otimes_{\mathbb{C}} \mathbb{C}[\mathbb{Z}/3\mathbb{Z}] \cong \mathbb{C}^2 \otimes \mathbb{C}^3.$$

This is not an artificial choice: for a finite abelian group $G \cong G_1 \times G_2$ with $\gcd(|G_1|, |G_2|) = 1$, the tensor decomposition $\mathbb{C}[G] \cong \mathbb{C}[G_1] \otimes \mathbb{C}[G_2]$ is a direct consequence of CRT [A]. Here $\otimes$ denotes the Hilbert space tensor product (not the algebraic tensor product): $\mathbb{C}[G]$ is equipped with the inner product $\langle f, g \rangle = |G|^{-1} \sum_{x \in G} f(x) \overline{g(x)}$, making the characters an orthonormal basis. The isomorphism $\mathbb{C}[G_1] \otimes \mathbb{C}[G_2] \cong \mathbb{C}[G]$ induced by $G_1 \times G_2 \cong G$ preserves this Hilbert space structure. Thus $\mathbb{C}^2 \otimes \mathbb{C}^3$ is a Hilbert space tensor product decomposition, and Theorem 3.7 applies directly. Note: here $\mathbb{C}[\mathbb{Z}/2\mathbb{Z}] \cong \mathbb{C}^2$ and $\mathbb{C}[\mathbb{Z}/3\mathbb{Z}] \cong \mathbb{C}^3$ are isomorphisms of Hilbert spaces (not of algebras); the character basis provides the orthonormal identification. By Theorem 3.7 ($a = 2$, $b = 3$), the gauge group of $p = 7$ is

$$G_7 = (U(1) \times \text{SU}(2) \times \text{SU}(3)) / \mathbb{Z}_6,$$

which is precisely the Standard Model gauge group (in its exact form with the central quotient removed). Identifying this mathematical symmetry group with the physical gauge group is a **constructive assumption** [B]. For the derivation of the Standard Model gauge group via noncommutative geometry, see Chamseddine–Connes [12] and [11].

3.4 Complete Table of Prime Channels

Remark 3.9. *In Table 1, “no splitting” means $(\mathbb{Z}/p\mathbb{Z})^*$ is cyclic and cannot be decomposed by CRT into two non-trivial factors, hence cannot produce two independent non-abelian gauge groups; in this case “spacetime dimension” is undefined. The column “spacetime dim.” is given by $\text{Herm}_a(\mathbb{C})$, where a is the order of the smallest prime-power CRT factor, and the dimension*

Table 1: Prime channels: the complete physics determined by each prime p

p	$\varphi(p)$	Spacetime dim.	$(\mathbb{Z}/p\mathbb{Z})^*$	Gauge group G_p	No. of chars.	Remarks
2	1	1	trivial	$\{1\}$	1	no internal stru
3	2	4	$\mathbb{Z}/2\mathbb{Z}$	$SU(2)$	2	$a = 2$, no SU
5	4	undefined	$\mathbb{Z}/4\mathbb{Z}$	no splitting	4	no CRT decomp
7	6	4	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$	$\mathbf{G}_{\text{SM}}$	6	our univers
11	10	4	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$	$SU(2) \times SU(5)$	10	$a = 2$, but SU
13	12	9	$\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$	$(U(1) \times SU(4) \times SU(3))/\mathbb{Z}_{12}$	12	$a = 3$, dim
17	16	undefined	$\mathbb{Z}/16\mathbb{Z}$	no splitting	16	no CRT decomp
19	18	4	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z}$	$SU(2) \times SU(9)$	18	$a = 2$, but SU

equals a^2 (Theorem 3.3). Note that multiple primes yield dimension 4 ($a = 2$); the uniqueness of $p = 7$ lies in the fact that CRT gives precisely $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ —the unique prime yielding the Standard Model gauge group $G_{\text{SM}} = (U(1) \times SU(2) \times SU(3))/\mathbb{Z}_6$ (Corollary 3.8). The column “No. of chars.” gives the number of characters modulo p , namely $\varphi(p) = p - 1$ (a purely arithmetic quantity). For $p = 7$, the $\varphi(7) = 6$ characters correspond to 6 fermion types; $6 = 3 \times 2$ can be divided into three generations each containing a particle and an antiparticle, but this identification is a physical interpretation [C] rather than a mathematical consequence.

Proposition 3.10 (Multiplicative independence of Euler factors [A]). *The Euler product $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$ gives a prime-by-prime decomposition of $\log \zeta(\beta)$:*

$$\log \zeta(\beta) = \sum_p \sum_{k=1}^{\infty} \frac{p^{-k\beta}}{k},$$

where the contributions of different primes p to $\log \zeta$, namely $\sum_k p^{-k\beta}/k$, are mutually independent—that is, $\log \zeta$ decomposes additively at the “prime-labeled frequencies $\log p$.”

Proof. $\log(1 - x)^{-1} = \sum_{k=1}^{\infty} x^k/k$ holds for $|x| < 1$. Set $x = p^{-\beta}$ ($|x| < 1$ for $\beta > 1$) and sum over all primes p . The contributions of distinct primes p, q are $\sum_k p^{-k\beta}/k$ and $\sum_k q^{-k\beta}/k$ respectively; as functions of β , they are linearly independent (since $\{p^{-\beta} : p \text{ prime}\}$ are linearly independent for distinct p). $\square$

4 Structure of the Multiverse

4.1 Coexistence: Complete Superposition at $\beta = 0$

Definition 4.1 (Multiverse superposition state [C]). *In the $\beta = 0$ limit, the KMS state of the BC system degenerates to the Haar measure—the uniform integral state on $\mathcal{A}_{\text{BC}}$. At this point all prime channels $p = 2, 3, 5, 7, 11, \dots$ coexist in equal-weight superposition, with no channel “selected.” We call this state the **multiverse superposition state**.*

Theorem 4.2 (Local decomposition of extremal KMS states [A]). *For $\beta > 1$, extremal KMS_{β} states are parametrized by $\widehat{\mathbb{Z}}^* = \prod_p \mathbb{Z}_p^*$. Each extremal state φ_{α} ($\alpha \in \widehat{\mathbb{Z}}^*$) is uniquely determined at each Euler factor by the image of α projected to $\mathbb{Z}_p^*$.*

Proof. This follows directly from Theorem 2.5(ii). $\widehat{\mathbb{Z}}^* = \prod_p \mathbb{Z}_p^*$ is the standard decomposition of the completed abelian group. $\square$

Remark 4.3 (Physical interpretation [C]). *Physically, $\beta > 1$ corresponds to “observation having occurred,” with a channel classically selected. The valuation decomposition of $\widehat{\mathbb{Z}}^*$ corresponds to the independent selection of each prime channel.*

4.2 Connections Between Channels

Proposition 4.4 (Composite numbers as inter-channel nodes [B]). *A composite number $n = p_1^{a_1} \cdots p_k^{a_k}$ belongs simultaneously to k prime channels. The Hamiltonian eigenvalue $\log n = \sum_i a_i \log p_i$ of the corresponding basis vector $|n\rangle$ in the BC algebra is a linear combination of the Hamiltonians of the individual channels, generating energy level correlations across channels.*

Definition 4.5 (Cross-channel projection [A]). *For primes p, q ($p \neq q$), define the **projection from the p -channel to the q -channel** as the image of the natural homomorphism $\pi_p : \mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}$ evaluated at q :*

$$\pi_p(q) = q \bmod p \in (\mathbb{Z}/p\mathbb{Z})^*.$$

More precisely, the value $\chi(q)$ of a Dirichlet character $\chi : (\mathbb{Z}/p\mathbb{Z})^ \rightarrow \mathbb{C}^*$ at q gives the “frequency response of the p -channel to the q -channel.” This is a one-way observation: the p -channel can only see $q \bmod p$, not q itself.*

Theorem 4.6 (ζ zeros as collective interference [C]). *The non-trivial zeros $\rho = \sigma + it$ of the Riemann ζ function satisfy the explicit formula [17, 16]:*

$$\psi(x) = x - \sum_{\zeta(\rho)=0} \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}),$$

*where $\psi(x) = \sum_{p^k \leq x} \log p$ is the von Mangoldt function. Each ζ zero contributes an oscillatory term with frequency t and decay rate $\sigma - 1/2$. Physical interpretation: the ζ zeros are the **collective interference pattern of all prime channels**.*

4.3 Time as a Channel Attribute

Proposition 4.7 (Time is a channel attribute [B]). *In the BC system, the “internal clock” frequency of each prime channel p is $\log p$, determined by the modular flow $\sigma_t(\mu_n) = n^{it} = e^{it \log n}$. The ratio of clock frequencies between distinct channels p and q ($p \neq q$) is $\log p / \log q$, which is irrational by the fundamental theorem of arithmetic; hence distinct prime channels are never synchronized—there is no global time.*

Proof. If $\log p / \log q = a/b \in \mathbb{Q}$ for positive integers a, b , then $p^b = q^a$. By the fundamental theorem of arithmetic, this is impossible when p and q are distinct primes. Therefore $\log p / \log q \notin \mathbb{Q}$, and the clocks of channels p and q share no common period. $\square$

Remark 4.8. *The “time” in Proposition 4.7 is the dynamical parameter t (the parameter of the modular flow σ_t , describing the internal oscillations of each channel). The β in the corollary below is a thermodynamic parameter (inverse temperature, describing the macroscopic evolution of the system). The two are independent in the BC system: t is local, β is global.*

Corollary 4.9 (β -axis is global [C]). *The inverse temperature β appears in the KMS conditions of all channels; it is a **global parameter**. All prime channels share the same β : they are born together ($\beta = 0$ superposition state), undergo the same phase transition ($\beta = 1$), and perish together ($\beta \rightarrow \infty$ heat death). β is the unique common clock spanning all channels.*

4.4 Cosmic Cycles

Theorem 4.10 (θ function modular transformation [A]). *The Jacobi θ function $\theta(x) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x}$ ($x > 0$) satisfies the modular transformation*

$$\theta\left(\frac{1}{x}\right) = \sqrt{x} \theta(x), \quad x > 0.$$

Remark 4.11 (Cosmic cycle interpretation [C]). *In the thermodynamic interpretation of the BC system, setting $x = e^\beta$, we get $1/x = e^{-\beta}$; this transformation connects $\beta = +\infty$ (heat death, ground state) with $\beta = 0$ (complete superposition, the starting point of the multiverse) into a single cycle: **heat death is the initial state of the next cosmic cycle.***

Remark 4.12 (Connection with Penrose’s CCC). *Penrose’s Conformal Cyclic Cosmology (CCC) [24] glues the asymptotic end state of one universe (eternal expansion) to the initial state of the next Big Bang via conformal rescaling. Theorem 4.10 provides an arithmetic version of the same idea: the θ modular transformation on the β -axis plays the role of the conformal rescaling in CCC. The difference is that CCC is geometric and depends on specific cosmological models, whereas the present version is algebraic and arises automatically from the structure of $\mathbb{Z}$.*

Proposition 4.13 (Galois selection and diversity [C]). *After each cosmic cycle, the BC ground state is parametrized by an element of $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q}) \cong \hat{\mathbb{Z}}^*$. Different Galois elements yield different “phase selections,” corresponding to different details (the different actions of the Galois group at the $\beta = 1$ symmetry-breaking point) under the same laws ($\mathbb{Z}$ invariant).*

Conjecture 4.14 (Galois diversity). *Different choices of $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$ give physically distinguishable universes, with differences manifested in the precise values of certain fundamental constants.*

4.5 Mirror Symmetry

Proposition 4.15 (Automorphisms of $\mathbb{Z}$ [A]). *$\text{Aut}(\mathbb{Z}, +) = \{\pm 1\}$ (only the identity and negation are automorphisms). The completed ξ function satisfies $\xi(s) = \xi(1-s)$, i.e., reflection symmetry about $\text{Re}(s) = 1/2$.*

Proof. $\text{Aut}(\mathbb{Z}, +) = \{\pm 1\}$ because an automorphism is determined by the image of 1, and 1 can only map to ± 1 (the generators). $\xi(s) = \xi(1-s)$ is the standard functional equation for the ζ function. $\square$

Remark 4.16 (Physical interpretation [C]). *$n \leftrightarrow -n$ corresponds to positive/negative energy or particle/antiparticle; $\xi(s) = \xi(1-s)$ ensures that the “anti-universe” is not an independent entity but the mirror image of the same universe.*

5 $p = 7$ —Our Universe

5.1 Three Independent Uniqueness Proofs

Theorem 5.1 (Uniqueness Theorem I: b_3 self-consistency equation [A*]). *The one-loop β -function coefficient of pure SU(3) gauge theory is $b_3 = 11$; upon including n_f generations of fermions it becomes*

$$b_3 = 11 - \frac{2n_f}{3}$$

(standard perturbative QCD [A]). Setting $n_f = \varphi(p) = p - 1$ (identifying all characters of $(\mathbb{Z}/p\mathbb{Z})^*$ with fermion generations—a **constructive assumption** [B]), and imposing the **self-consistency condition** that the β -function coefficient of $SU(3)$ equals the prime defining the gauge group, i.e., $b_3 = p$ [C], the equation

$$11 - \frac{2(p-1)}{3} = p$$

has the unique integer solution $p = 7$ (algebraic derivation [A]).

Proof. (Algebraic derivation [A].) Expanding: $11 - \frac{2p-2}{3} = p$, i.e., $33 - 2p + 2 = 3p$, so $5p = 35$, giving $p = 7$. $\square$

Theorem 5.2 (Uniqueness Theorem II: four-dimensional Minkowski spacetime + $SU(3)$ [A]). *Among all primes p , $p = 7$ is the unique prime simultaneously satisfying the following two conditions:*

- (i) *spacetime dimension = 4 (the minimal prime-power CRT factor has order $a = 2$, i.e., $\mathbb{Z}/2\mathbb{Z}$ is a CRT direct summand; then $\text{Herm}_2(\mathbb{C})$ yields dimension 4, by Theorem 3.3);*
- (ii) *the gauge group contains the $SU(3)$ color group (CRT has a $\mathbb{Z}/3\mathbb{Z}$ factor, not $\mathbb{Z}/9\mathbb{Z}$ or higher powers of 3).*

Proof. The full derivation of condition (i) (spacetime dimension = 4) is given in [4] (Theorem 3.3). The argument for condition (ii) (CRT contains $\mathbb{Z}/3\mathbb{Z}$) is given in [2] (Theorem 3.6). Combining both conditions: $v_2(p-1) = 1$ and $v_3(p-1) = 1$, i.e., $6 \mid (p-1)$ but $4 \nmid (p-1)$ and $9 \nmid (p-1)$. Writing $p-1 = 6m$ with $\gcd(m, 6) = 1$, $m = 1$ gives $p = 7$; for $m \geq 5$ the CRT decomposition has more than two factors, and Theorem 3.7 (see [2]) does not directly apply. For $p = 31$: $(\mathbb{Z}/31\mathbb{Z})^* \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$ has three CRT factors; the tensor product $\mathbb{C}^2 \otimes \mathbb{C}^3 \otimes \mathbb{C}^5$ does not yield the Standard Model gauge group (the additional $SU(5)$ factor is absent in nature). Hence $p = 31$ is excluded. Hence $p = 7$ is unique. $\square$

Theorem 5.3 (Uniqueness Theorem III: CRT gauge group [A]). *Among all primes p , $(\mathbb{Z}/p\mathbb{Z})^* \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ (permitting the $SU(2) \times SU(3)$ decomposition) holds if and only if $p = 7$.*

Proof. $p-1 = |\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}| = 6$, so $p = 7$. See [2] for details. For any other prime q : if $q-1 = 6k$ ($k > 1$), then $(\mathbb{Z}/q\mathbb{Z})^* \cong \mathbb{Z}/(q-1)\mathbb{Z}$ is not isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ (the latter has order 6, while $q-1 > 6$). If $q-1 \not\equiv 0 \pmod{6}$, then $(\mathbb{Z}/q\mathbb{Z})^*$ does not contain the required subgroup structure. $\square$

Theorem 5.4 (Uniqueness Theorem IV: three generations of fermions [A]). *$(p-1)/2 = 3$ (exactly three H_2 -doublet orbits) holds if and only if $p = 7$. For $p = 11$: $(p-1)/2 = 5$ (too many generations); for $p = 13$: $(p-1)/2 = 6$.*

Proof. $(p-1)/2 = 3 \Rightarrow p = 7$. $\checkmark$ $\square$

Theorem 5.5 (Uniqueness Theorem V: geometric–arithmetic coincidence [A]). *$(2\varphi_2^{(p)-2}) = \varphi(p) \cdot p + \varphi(d_1) + \varphi(d_2)$ holds if and only if $p = 7$: $\binom{10}{2} = 45 = 42 + 3$. For $p = 11$: $\binom{18}{2} = 153 \neq 115$; for $p = 13$: $\binom{22}{2} = 231 \neq 160$. That is, $\dim \text{Adj } SO(2\varphi(p) - 2)$ (geometric route) = $|\text{Aff}(\mathbb{F}_p)| + \sum \varphi(d_i)$ (arithmetic route) holds only for $p = 7$.*

Proof. Direct computation. $p = 7$: $\binom{10}{2} = 45$, $6 \times 7 + 1 + 2 = 45$. $\checkmark$

$p = 11$: $\binom{18}{2} = 153$, $10 \times 11 + 1 + 4 = 115$. $153 \neq 115$. $\checkmark$

$p = 13$: $\binom{22}{2} = 231$, $12 \times 13 + 2 + 2 = 160$. $231 \neq 160$. $\checkmark$ $\square$

Corollary 5.6 (Physical consequences of $p = 7$ [A]). *From $p = 7$ one can derive:*

- (i) $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, minimal factor $a = 2$, $\dim_{\mathbb{R}} \text{Herm}_2(\mathbb{C}) = 4$, Minkowski signature $(1, 3)$ (four-dimensional spacetime, Theorem 3.3);
- (ii) $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \Rightarrow$ gauge group $U(1) \times \text{SU}(2) \times \text{SU}(3)$ (containing $U(1)$, hence electromagnetism);
- (iii) $\varphi(7) = 6$ characters $\Rightarrow 3$ generations $\times 2$ (particle/antiparticle) fermions;
- (iv) $p - 2 = 5 = \beta \Rightarrow$ thermodynamic determination of coupling constants.

5.2 Uniqueness of $p = 7$: Anthropic Principle and Uniqueness

Proposition 5.7 (No fine-tuning [B]). *It is commonly believed that physical constants are “fine-tuned” to values permitting the existence of life [20, 21]. In the framework of this paper, $p = 7$ is the **unique** prime simultaneously admitting the following conditions:*

- four-dimensional spacetime ($\text{Herm}_2(\mathbb{C})$, i.e., CRT minimal factor $a = 2$, $\dim_{\mathbb{R}} = 4$, Minkowski signature $(1, 3)$);
- $\text{SU}(2) \times \text{SU}(3)$ gauge structure (CRT decomposes as $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$);
- three generations of fermions ($\varphi(7)/2 = 3$).

Therefore “why $p = 7$ ” and “why observers exist” are two formulations of the same question, and the answer is: $p = 7$ is the unique prime capable of producing observers. This is not fine-tuning but a unique solution. Unlike the landscape approach [21] (universes selected from an enormous ensemble via anthropic reasoning), here the arithmetic structure of $\mathbb{Z}$ singles out $p = 7$ as the unique possibility.

5.3 Further Research Directions

The main computational directions opened by the $p = 7$ framework—most of which have already been carried out in companion papers—are as follows:

- **Quantum mechanics as theorems [resolved]:** The C^* -positivity of the BC algebra yields a Hilbert space via the GNS construction; the Born rule $P = |c_n|^2$ is uniquely determined by the Cauchy equation and non-negative squeezing [A]; tensor products follow directly from the unique factorization theorem. See [1].
- **Unification of four forces [resolved]:** The factor-preserving unitary group of the CRT decomposition $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ gives exactly $G_{\text{SM}} = (U(1) \times \text{SU}(2) \times \text{SU}(3))/\mathbb{Z}_6$ [A]; the KMS modular Hamiltonian and the von Mangoldt explicit formula give matter–geometry duality [A]. See [2].
- **Standard particle mass spectrum [resolved]:** The six fermion sectors are uniquely determined by the six characters [A]; the Jacobi sum phase δ yields the Koide formula $Q = 2/3$ [A], all three CKM entries to within $< 2.5\%$ [A*]; neutrino mass ratio $R = 33.0$ (deviation 1.5%) [A*], $\delta_{CP} = -90^\circ$ [A]. See [3, 5].
- **Cosmological constant [resolved]:** BC positive curvature $\kappa(n) > 0$ gives $\Lambda > 0$ [A*]; the quantitative derivation gives $\Lambda = 2.86 \times 10^{-122}$ (Planck units, deviation 0.4%) [A*]. See [4].
- **Motivic interpretation [still open]:** How the prime channel structure relates to motivic theory and the theory of L -functions in arithmetic geometry; see Deninger [13] on motivic local L -factors, and Connes–Marcolli [10] on renormalization and the motivic Galois group.

6 Cross-Channel Physics—Dark Matter and Gravity

6.1 Dark Matter = Mass from Other Channels

Proposition 6.1 (Arithmetic origin of dark matter [C]). *Observers within the $p = 7$ channel can only detect gauge interactions of the $p = 7$ channel ($U(1) \times \text{SU}(2) \times \text{SU}(3)$). Matter in other channels $p \neq 7$ shares the β -axis with matter in the $p = 7$ channel (gravity arises from the global nature of β), but does not participate in the gauge interactions of the $p = 7$ channel (gauge groups of different channels are mutually independent). Consequently, matter from other prime channels manifests in the $p = 7$ channel as:*

- gravitational interaction (β is global—the arithmetic realization of gravity is given by the BC modular Hamiltonian $K = \beta H + \log \zeta(\beta) \cdot I[A]$ and the von Mangoldt matter–geometry identity [A], detailed in [2]);
- no electromagnetic force (channels $p \neq 7$ do not carry $U(1)_{p=7}$);
- no strong force (the $\text{SU}(3)$ of $p \neq 7$ channels differs from that of $p = 7$).

This is precisely the observational signature of dark matter.

Remark 6.2 (Dark matter fraction). *The qualitative prediction—dark matter originates from other prime channels—is a structural consequence of the multiverse framework. Quantitatively deriving $\Omega_{\text{DM}}/\Omega_b$ from the BC algebra remains an open problem: the correct weight formula $w(p)$ for prime channels has not yet been established from first principles.*

Conjecture 6.3 (Arithmetic expression for the dark matter ratio).

$$\frac{\Omega_{\text{DM}}}{\Omega_b} = (p - 2) + \frac{\varphi(p - 1)}{p - 1} = 5 + \frac{1}{3} = \frac{16}{3} \approx 5.333. \quad (1)$$

Observed value 5.364, deviation 0.58%. The origin and proof level of each term are as follows:

- **Main term** $(p - 2) = 5$ [A]: the $\varphi(7) - 1 = 5$ non-trivial characters of $(\mathbb{Z}/7\mathbb{Z})^*$ have no pole at $s = 1 \rightarrow$ do not participate in the $\beta = 1$ phase transition $\rightarrow$ invisible; Galois equivalence ensures equal energy density across sectors.
- **Correction term** $\varphi(p - 1)/(p - 1) = \varphi(6)/6 = 2/6 = 1/3$ [A*]: $\varphi(p - 1) = |(\mathbb{Z}/(p - 1)\mathbb{Z})^*|$ equals the number of automorphisms of the character space under charge conjugation $\chi \mapsto \bar{\chi}$. The 5 dark sectors form 2 conjugate pairs $(\chi_1 \leftrightarrow \chi_5, \chi_2 \leftrightarrow \chi_4) + 1$ self-conjugate $(\chi_3, \text{the Legendre symbol})$. The conjugation density $\varphi(p - 1)/(p - 1)$ is consistent with the observed correction; the precise KMS dynamical derivation remains to be completed.

Remark 6.4 (Dark matter from a multiverse perspective [A/B]). *In the $p = 7$ channel, 1 of the 6 characters of $(\mathbb{Z}/7\mathbb{Z})^*$ (the trivial χ_0) corresponds to visible matter, while 5 correspond to invisible sectors. The multi-component structure of dark matter is determined by charge conjugation: 2 Dirac-type pairs $(\chi_k \leftrightarrow \bar{\chi}_k) + 1$ Majorana-type $(\chi_3 = \bar{\chi}_3)$. This constitutes a testable prediction: dark matter should consist of at least two species (see §7 for details).*

6.2 CP Violation = Distortion of Channel Projection

Proposition 6.5 (Arithmetic origin of CP violation [B]). *At the level of $\mathbb{Z}$, the map $n \mapsto -n$ gives perfect CP symmetry (Proposition 4.15). When $\mathbb{Z}$ is projected onto the $p = 7$ channel, the characters $\chi : (\mathbb{Z}/7\mathbb{Z})^* \rightarrow \mathbb{C}^*$ are complex-valued functions; their conjugates $\bar{\chi}$ correspond to “antiparticle channels.” The phase difference between χ and $\bar{\chi}$ is non-zero (when χ is non-real), thereby producing CP violation in the $p = 7$ channel. Specifically, the 4 non-real characters modulo 7 each contribute a phase; their sum is non-zero, yielding CP violation in the $p = 7$ channel, while the degree of CP violation differs in other channels.*

Remark 6.6. *This mechanism provides an arithmetic explanation for CP violation: it originates from the phase loss incurred when projecting the perfect mirror symmetry of $\mathbb{Z}$ onto a specific prime channel, rather than from some “exogenous” parameter.*

6.3 Riemann Hypothesis = Harmony of the Multiverse

Theorem 6.7 (Multiverse interpretation of RH [C]). *The Riemann Hypothesis (RH) is equivalent to the statement that all non-trivial zeros of ζ lie on the critical line $\Re(s) = 1/2$. In the multiverse picture (Theorem 4.6), the ζ zeros are the collective interference pattern of all prime channels. Zeros on the critical line mean that the “decay rate” of each interference pattern is exactly 0 ($\sigma - 1/2 = 0$), i.e., all channels are **perfectly balanced**. If a zero deviates from the critical line ($\sigma \neq 1/2$), then certain channels’ interference grows exponentially (when $\sigma > 1/2$) or decays (when $\sigma < 1/2$), corresponding to a systematic imbalance among channels.*

Conjecture 6.8 (Physical necessity of RH [C]). *If the multiverse system is in a stable (KMS equilibrium) state, then there is no systematic imbalance among channels, and hence RH holds. That is: **multiverse exists (has a KMS state)** $\Rightarrow$ **RH**.*

A precise mathematical formulation of this conjecture requires establishing a rigorous connection between the condition of “stable multiverse” and the distribution of ζ zeros; see Connes [8] on the connection between RH and operator spectra, and Deninger [14] on the dynamical systems approach.

6.4 Algebraic Meaning of Anthropic Selection

Proposition 6.9 (Unique observable channel [A]). *Suppose an “observer” requires all of the following conditions to hold simultaneously:*

- *spacetime dimension $d_s \geq 4$ (gravity not too weak, molecules can exist stably);*
- *existence of a non-abelian gauge group (enabling fermions to acquire mass, making chemistry possible);*
- *existence of three generations of fermions (CKM/PMNS matrices are non-trivial, CP violation is sufficiently large).*

Then by Table 1, Theorem 5.2, and Theorem 5.3, the unique prime p satisfying all three conditions simultaneously is $p = 7$.

7 Mathematics as a Cross-Channel Telescope

7.1 The Limitations of Physics

Proposition 7.1 (Invisibility within a channel [A]). *Any physical experiment within the $p = 7$ channel cannot directly probe the gauge interactions of other prime channels, because the gauge groups of different channels are mutually independent (Proposition 3.10). Specifically, accelerators probe $SU(2) \times SU(3)$ charges, which have no meaning in other prime channels.*

7.2 The Power of Mathematics

Theorem 7.2 (Mathematics is the unique cross-universe telescope [C]). *$\mathbb{Z}$ is the common root of all prime channels—the structure of $\mathcal{A}_{BC}$ holds simultaneously for all channels. Therefore, by studying the mathematical properties of $\mathbb{Z}$ (the ζ function, characters, L-functions, etc.), one can infer the existence of other channels and their physical properties, even if those channels are never directly observed. **Mathematics is the unique telescope capable of seeing other channels.***

Remark 7.3. *This statement is not a metaphor but a proposition with mathematical content: the CRT decompositions and gauge group structures of the $p = 2, 3, 5, 11, \dots$ channels in Table 1 are rigorously derived from the algebraic structure of $\mathcal{A}_{BC}$ [A]; the spacetime dimension column (given by $\text{Herm}_a(\mathbb{C})$) is likewise an algebraic property [A] (Theorem 3.3). Whether they are physical reality cannot currently be verified experimentally; but their mathematical existence (as local factors of the BC algebra) is unconditional. The role of number theory and L-functions as a “dictionary” between different mathematical universes is systematically developed in [15, 16, 13].*

7.3 Time and Its Relation to $\mathbb{Z}$

Proposition 7.4 ($\mathbb{Z}$ is outside of time [C]). *The modular flow σ_t is the dynamics of the BC algebra; it **acts on** elements of $\mathcal{A}_{BC}$, whereas $\mathbb{Z}$ itself (as the initial object) is the foundation of this dynamics and is not subject to its action. In physical language: time (σ_t) is a structure internal to the universe ($\mathcal{A}_{BC}$), while the foundation of the universe ($\mathbb{Z}$) is itself timeless. **The structure of $\mathbb{Z}$ precedes the concept of time.***

Remark 7.5 ($\mathbb{Z}$ is the root). *$\mathbb{Z}$ is the initial object in the category of rings (Definition 1.1), logically prior to any specific ring. It does not inhabit any particular physical channel; it is the mathematical prerequisite for all channels.*

7.4 Testable Predictions

The multiverse framework yields the following testable predictions:

- P1. Dark matter is multi-component [A*].** The 5 invisible character sectors form under charge conjugation: 2 conjugate pairs ($\chi_1 \leftrightarrow \chi_5, \chi_2 \leftrightarrow \chi_4$) + 1 self-conjugate (χ_3). Conjugate pairs correspond to Dirac-type dark matter (particle $\neq$ antiparticle); the self-conjugate corresponds to Majorana-type dark matter (particle = antiparticle). **Prediction:** dark matter consists of at least two species. Testable: future DM direct detection experiments finding a multi-component signal.
- P2. Dark matter is absolutely stable [A].** The gauge groups of different character sectors are completely independent (Proposition 3.10), with no cross-sector mixing $\rightarrow$ dark matter cannot decay into visible matter. **Prediction:** no dark matter decay signal exists.
- P3. Five dark sectors have equal energy density [A].** Galois equivalence ensures each dark character sector contributes the same energy density. **Prediction:** Ω_{DM} consists of 5 equal parts, each $\approx \Omega_{DM}/5 \approx 0.053$.
- P4. Dark matter does not self-annihilate [A*].** Particles and antiparticles of different prime channels belong to different character sectors. Annihilation within the same channel is possible, but cross-channel annihilation is forbidden by gauge group independence (Proposition 3.10). **Prediction:** no large-scale dark matter self-annihilation signal exists. Consistent with current experiments: Fermi-LAT has not detected confirmed dark matter annihilation gamma rays.
- P5. Gravitational wave background has discrete frequency structure [C].** Collective fluctuations of the BC system manifest, in the continuum approximation, as a stochastic background signal with discrete frequency structure. The characteristic frequencies of different prime channels are $\omega_p = \log p$; cross-channel coupling (via the global nature of the $\beta = 1$ pole) produces enhancements in the gravitational wave background at frequency spacing ratios $\log 3/\log 2, \log 5/\log 2, \log 7/\log 2$. **Prediction:** the stochastic gravitational wave background has discrete structure at prime frequency ratios. Testable: LISA (~ 2035) and Pulsar Timing Arrays (PTA, currently operating).

7.5 Open Problems

- Q1. Galois distinguishability:** Are different elements of $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$ physically distinguishable “instances of the multiverse”? Are their differences reflected in the special values of certain L -functions?
- Q2. Cross-channel gravity:** Can the physical field corresponding to the global nature of β be perceived within the $p = 7$ channel as “extra dimensions”? What is the precise connection with Kaluza–Klein theory?
- Q3. Dark matter weights:** The precise definition of the weight $w(p)$ in Conjecture 6.3, and a scheme for rigorously deriving $\Omega_{\text{DM}}/\Omega_b$ from the BC algebra.
- Q4. Physical proof of RH:** Can RH be rigorously derived from the physical assumption of “multiverse stability” (making Conjecture 6.8 precise)?
- Q5. Rhythm of cosmic cycles:** Does the θ modular transformation (Theorem 4.10) correspond to a detectable signal between “Aeons” in Penrose’s CCC, potentially visible in the CMB?

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